Biochemistry Cume
April 25, 2003

Provide answers to four of the five questions below.

1. (25 Points) A newly purified protein elutes from a gel permeation column as a single band with a molecular weight of 120,000 under low salt conditions. In the presence of high salt, the protein elutes as two bands - one having a molecular weight of 100,000 and the other of 10,000. SDS gel electrophoresis of the 120,000 molecular weight form of the protein also gives bands of 100,000 and 10,000 in the absence of mercaptoethanol. However, it done in the presence of mercaptoethanol, bands of 50,000 and 10,000 are observed. Based on these observations, describe as much as you can about the structure of the protein.

2. (25 points) Consider the equilibrium for the dissociation of a weak acid

$$HA \Longleftrightarrow A + H^+$$

Show that for such a reaction, $\Delta G^o = RT(pKa)$.

Now consider the following reaction (for an amino acid with no ionizible groups in R-group)

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      COOH                          COO-
       |                             |
NH2-C-H     <==========>     +NH3-C-H
       |                             |
       R                             R
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For this reaction, show that $\Delta G^o = RT\ (pKa1 - pKa2)$ where pKa1 is the pKa of the carboxyl group and pKa2 is the pKa of the amino group.

3. (25 points) You are using a buffer having an initial pH of 7.3 and a pKa of 7.0 is the study of a biochemical reaction. At the end of the reaction, you measure the pH and find it to be 6.7. What would the final pH have been had there been no buffer present?

4. (25 Points) A reaction of the type

$$A + B \Longleftrightarrow 2\,C$$

is started with (A), (B), and (C) being 4, 3, and 1 mM, respectively. The concentration of C is measured after the reaction has been allowed to reach equilibrium and found to be 3 times the initial concentration. Determine ΔG for the reaction which has occurred.

5. (25 points) You have a mixture of four proteins with the following characteristics

	Molecular weight	Charge
A	30,000	+10
B	30,000	-10
C	10,000	+10
D	10,000	-10

None of these proteins are known to have any specific interaction with one another or with any other biological molecule.

Develop a strategy involving only chromatographic techniques which you could utilize to obtain each of these proteins in pure form. Indicate in your strategy the methods you would use, the order in which you would use them, and which proteins would be obtained at which steps.